

Inner universe

simulated observer records

particles, clocks, screens, measurements

U_{OPH} in $Fix(T)$

P^* local pixel identity

α field-width readback

closure coordinates

not knobs, not fitted constants

the loop only survives at the fixed points

N^* total screen capacity

$\mathcal{A}$ horizon closure

Outer universe

simulating readback rule

repair, select, return the same records

$T(U_{OPH}) = U_{OPH}$